# CPS_DATA_EXERCISE

2025-12-07

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## Setting Up the Environment

## Data Wrangling and Analysis

### (3) Importing Data

```r
CPSDATA <- read.csv("CPSDATA.csv")
str(CPSDATA)
```

```
'data.frame':   1331621 obs. of  22 variables:
 $ YEAR       : int  2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
 $ SERIAL     : int  1 4 5 5 5 6 6 8 9 9 ...
 $ MONTH      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ HWTFINL    : num  2556 2482 3032 3032 3032 ...
 $ CPSID      : num  2.02e+13 2.02e+13 2.02e+13 2.02e+13 2.02e+13 ...
 $ ASECFLAG   : int  NA NA NA NA NA NA NA NA NA NA ...
 $ ASECWTH    : num  NA NA NA NA NA NA NA NA NA NA ...
 $ REGION     : int  32 32 32 32 32 32 32 32 32 32 ...
 $ STATEFIP   : int  1 1 1 1 1 1 1 1 1 1 ...
 $ STATECENSUS: int  63 63 63 63 63 63 63 63 63 63 ...
 $ COUNTY     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ PERNUM     : int  1 1 1 2 3 1 2 1 1 2 ...
 $ WTFINL     : num  2556 2482 3032 2278 1697 ...
 $ CPSIDP     : num  2.02e+13 2.02e+13 2.02e+13 2.02e+13 2.02e+13 ...
 $ CPSIDV     : num  2.02e+14 2.02e+14 2.02e+14 2.02e+14 2.02e+14 ...
 $ ASECWT     : num  NA NA NA NA NA NA NA NA NA NA ...
 $ AGE        : int  58 64 44 80 35 23 35 60 58 58 ...
 $ SEX        : int  2 2 1 2 2 2 1 1 1 2 ...
 $ RACE       : int  100 100 100 100 100 100 100 200 200 200 ...
 $ MARST      : int  4 4 4 5 6 6 4 6 1 1 ...
 $ EDUC99     : int  10 15 7 5 10 11 10 10 10 14 ...
 $ INCWAGE    : int  NA NA NA NA NA NA NA NA NA NA ...
```

```r
CPSDATA[] <- lapply(CPSDATA, function(x) {
  if (is.character(x) && all(grepl("^[0-9]+$", x))) {
    as.numeric(x)
```

```
  } else {
    x
  }
})
```

## (4) SEX Variable

```
CPSDATA$FEMALE <- ifelse(CPSDATA$SEX == 2, 1, 0)
head(CPSDATA$FEMALE)
```

```
[1] 1 1 0 1 1 1
```

## (5) RACE Variable

```
CPSDATA$NONWHITE <- ifelse(CPSDATA$RACE != 100 & CPSDATA$RACE != 999, 1, 0)
head(CPSDATA$NONWHITE)
```

```
[1] 0 0 0 0 0 0
```

## (6) MARST Variable

```
CPSDATA$MARRIED <- ifelse(CPSDATA$MARST == 1, 1, 0)
head(CPSDATA$MARRIED)
```

```
[1] 0 0 0 0 0 0
```

## (7) EDUC99 Variable

```r
CPSDATA$HIGHSCHOOL <- ifelse(CPSDATA$EDUC99 >= 10, 1, 0)
head(CPSDATA$HIGHSCHOOL)
```

```
[1] 1 1 0 0 1 1
```

## (8) INCWAGE Variable

```r
CPSDATA$INCWAGE[CPSDATA$INCWAGE %in% c(99999998, 99999999)] <- NA
head(CPSDATA$INCWAGE)
```

```
[1] NA NA NA NA NA NA
```

## (9) REGION Variable

```r
CPSDATA$REGION <- ifelse(CPSDATA$REGION %in% c(11, 12), 1,
                         ifelse(CPSDATA$REGION %in% c(21, 22), 2,
                         ifelse(CPSDATA$REGION %in% c(31, 32, 33), 3, 4)))
head(CPSDATA$REGION)
```

```
[1] 3 3 3 3 3 3
```

## (10) Droping Old Variables

```r
CPSDATA <- subset(CPSDATA, select = -c(SEX, RACE, MARST, EDUC99))
head(CPSDATA)
```

```
  YEAR SERIAL MONTH  HWTFINL       CPSID ASECFLAG ASECWTH REGION STATEFIP
1 2024      1     1 2555.891 2.02401e+13       NA      NA      3        1
2 2024      4     1 2481.663 2.02401e+13       NA      NA      3        1
3 2024      5     1 3032.100 2.02401e+13       NA      NA      3        1
4 2024      5     1 3032.100 2.02401e+13       NA      NA      3        1
```

```
5 2024      5     1 3032.100 2.02401e+13       NA      NA      3        1
6 2024      6     1 2347.767 2.02401e+13       NA      NA      3        1
  STATECENSUS COUNTY PERNUM   WTFINL      CPSIDP      CPSIDV ASECWT AGE INCWAGE
1          63      0      1 2555.891 2.02401e+13 2.02401e+14     NA  58      NA
2          63      0      1 2481.663 2.02401e+13 2.02401e+14     NA  64      NA
3          63      0      1 3032.100 2.02401e+13 2.02401e+14     NA  44      NA
4          63      0      2 2278.490 2.02401e+13 2.02401e+14     NA  80      NA
5          63      0      3 1697.057 2.02401e+13 2.02401e+14     NA  35      NA
6          63      0      1 2347.767 2.02401e+13 2.02401e+14     NA  23      NA
  FEMALE NONWHITE MARRIED HIGHSCHOOL
1      1        0       0          1
2      1        0       0          1
3      0        0       0          0
4      1        0       0          0
5      1        0       0          1
6      1        0       0          1
```

## (11) Descriptive Statistics

```r
library(dplyr)
continuous_vars <- CPSDATA %>%
  select(INCWAGE) %>%
  summarise(
    Mean = mean(INCWAGE, na.rm = TRUE),
    SD = sd(INCWAGE, na.rm = TRUE),
    Min = min(INCWAGE, na.rm = TRUE),
    Max = max(INCWAGE, na.rm = TRUE)
  )

continuous_vars <- data.frame(
  Statistic = c("Mean", "SD", "Min", "Max"),
  INCWAGE = c(continuous_vars$Mean,
              continuous_vars$SD,
              continuous_vars$Min,
              continuous_vars$Max)
)

print("Descriptive statistics for continuous variables:")
```

```
[1] "Descriptive statistics for continuous variables:"
```

```r
print(continuous_vars)
```

```
  Statistic   INCWAGE
1      Mean   40807.3
2        SD   71569.5
3       Min       0.0
4       Max 1399999.0
```

```r
categorical_vars <- CPSDATA %>%
  select(FEMALE, NONWHITE, MARRIED, HIGHSCHOOL, REGION)

categorical_stats <- lapply(names(categorical_vars), function(var) {
  freq_table <- table(categorical_vars[[var]], useNA = "ifany")
  prop_table <- prop.table(freq_table)

  data.frame(
    Variable = var,
    Category = names(freq_table),
    Frequency = as.numeric(freq_table),
    Proportion = round(as.numeric(prop_table), 4)
  )
})

categorical_stats_df <- do.call(rbind, categorical_stats)

print("Descriptive statistics for categorical variables:")
```

```
[1] "Descriptive statistics for categorical variables:"
```

```r
print(categorical_stats_df)
```

```
     Variable Category Frequency Proportion
1      FEMALE        0    650469     0.4885
2      FEMALE        1    681152     0.5115
3    NONWHITE        0   1055833     0.7929
```

```
4    NONWHITE        1    275788     0.2071
5     MARRIED        0    779509     0.5854
6     MARRIED        1    552112     0.4146
7  HIGHSCHOOL        0    365356     0.2744
8  HIGHSCHOOL        1    966265     0.7256
9      REGION        1    209784     0.1575
10     REGION        2    261771     0.1966
11     REGION        3    491238     0.3689
12     REGION        4    368828     0.2770
```

## (12) INCWAGE Fractions

```
CPSDATA$INCWAGE <- ifelse(CPSDATA$INCWAGE == 0, 0.01, CPSDATA$INCWAGE)
```

## (13) Model 1

```
library(sandwich)
library(lmtest)
model1 <- lm(log(INCWAGE) ~ FEMALE + NONWHITE + MARRIED + HIGHSCHOOL,
             data = CPSDATA)

coeftest(model1, vcov = vcovHC(model1, type = "HC"))
```

```
t test of coefficients:

             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  1.040703   0.058841  17.6868 < 2.2e-16 ***
FEMALE      -1.575856   0.042757 -36.8558 < 2.2e-16 ***
NONWHITE    -0.136091   0.050671  -2.6858  0.007237 **
MARRIED      0.821394   0.044216  18.5770 < 2.2e-16 ***
HIGHSCHOOL   4.626828   0.058298  79.3647 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## (14) Model 2

```r
model2 <- lm(log(INCWAGE) ~ FEMALE + NONWHITE + FEMALE:NONWHITE +
               MARRIED + HIGHSCHOOL, data = CPSDATA)

coeftest(model2, vcov = vcovHC(model2, type = "HC"))
```

```
t test of coefficients:

                 Estimate Std. Error  t value  Pr(>|t|)
(Intercept)      1.120801   0.060000  18.6800 < 2.2e-16 ***
FEMALE          -1.734323   0.048850 -35.5028 < 2.2e-16 ***
NONWHITE        -0.496315   0.072996  -6.7992 1.057e-11 ***
MARRIED          0.821851   0.044211  18.5893 < 2.2e-16 ***
HIGHSCHOOL       4.628196   0.058276  79.4189 < 2.2e-16 ***
FEMALE:NONWHITE  0.682638   0.100893   6.7659 1.331e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## (15) Model 3

```r
CPSDATA$R2 <- ifelse(CPSDATA$REGION == 2, 1, 0)
CPSDATA$R3 <- ifelse(CPSDATA$REGION == 3, 1, 0)
CPSDATA$R4 <- ifelse(CPSDATA$REGION == 4, 1, 0)

model3 <- lm(log(INCWAGE) ~ FEMALE + NONWHITE + FEMALE:NONWHITE +
               MARRIED + HIGHSCHOOL + R2 + R3 + R4, data = CPSDATA)

coeftest(model3, vcov = vcovHC(model3, type = "HC"))
```

```
t test of coefficients:

                 Estimate Std. Error  t value  Pr(>|t|)
```

```
(Intercept)      1.305934   0.078722  16.5893 < 2.2e-16 ***
FEMALE          -1.731982   0.048804 -35.4882 < 2.2e-16 ***
NONWHITE        -0.424843   0.073132  -5.8093 6.290e-09 ***
MARRIED          0.825217   0.044179  18.6787 < 2.2e-16 ***
HIGHSCHOOL       4.608450   0.058277  79.0784 < 2.2e-16 ***
R2               0.345342   0.073229   4.7159 2.409e-06 ***
R3              -0.490196   0.064792  -7.5656 3.888e-14 ***
R4              -0.237274   0.067413  -3.5197 0.0004322 ***
FEMALE:NONWHITE  0.689650   0.100845   6.8387 8.029e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## (16) Dummification of "Region" and Explanation

Excluding the first region avoids perfect multicollinearity, commonly known as the "dummy variable trap". By excluding one region, the model can estimate the effects of the other regions relative to the excluded region and serve as a comparison.

## (17) Table Presentation of the Results

```r
library(stargazer)
stargazer(model1, model2, model3,
          type = "text",
          title = "Regression Results",
          column.labels = c("Model 1", "Model 2", "Model 3"),
          covariate.labels = c("FEMALE", "NONWHITE", "FEMALE:NONWHITE", "MARRIED", "HIGHSCHOOL", "R2", "R3", "R4"),
          dep.var.labels = "Log(INCWAGE)")
```

```
Regression Results
==============================================================================================================
                                                       Dependent variable:
                    ------------------------------------------------------------------------------------------
                                                          Log(INCWAGE)
                                Model 1                      Model 2                      Model 3
                                  (1)                          (2)                          (3)
--------------------------------------------------------------------------------------------------------------
```

| | | | |
|---|---|---|---|
| FEMALE | -1.576*** | -1.734*** | -1.732*** |
| | (0.043) | (0.049) | (0.049) |
| NONWHITE | -0.136*** | -0.496*** | -0.425*** |
| | (0.051) | (0.074) | (0.074) |
| FEMALE:NONWHITE | 0.821*** | 0.822*** | 0.825*** |
| | (0.044) | (0.044) | (0.044) |
| MARRIED | 4.627*** | 4.628*** | 4.608*** |
| | (0.062) | (0.062) | (0.062) |
| HIGHSCHOOL | | | 0.345*** |
| | | | (0.074) |
| R2 | | | -0.490*** |
| | | | (0.065) |
| R3 | | | -0.237*** |
| | | | (0.067) |
| R4 | | 0.683*** | 0.690*** |
| | | (0.101) | (0.101) |
| Constant | 1.041*** | 1.121*** | 1.306*** |
| | (0.062) | (0.063) | (0.081) |
| Observations | 115,836 | 115,836 | 115,836 |
| R2 | 0.066 | 0.066 | 0.068 |
| Adjusted R2 | 0.066 | 0.066 | 0.067 |
| Residual Std. Error | 7.275 (df = 115831) | 7.273 (df = 115830) | 7.267 (df = 115827) |
| F Statistic | 2,031.495*** (df = 4; 115831) | 1,634.899*** (df = 5; 115830) | 1,048.518*** (df = 8; 115827) |

Note: *p<0.1; **p<0.05; ***p<0.01

## (17.1) Interpretation of Results from Model 3

The regression analysis (Model 3) examines the effects of gender, race, marital status, education, and region on the natural logarithm of income (`log(INCWAGE)`) using heteroskedasticity-robust standard errors. The model includes interaction terms and regional dummy variables to account for potential differences across groups and regions.

## (17.2) Main Findings:

1. **Gender**:
   - The coefficient for `FEMALE` is significant ($\beta$ = -1.732, SE = 0.049, p < .01), indicating that women, on average, had 173.2% lower income in 2024 compared to men, holding all other variables constant.
2. **Race**:
   - The coefficient for `NONWHITE` is significant ($\beta$ = -0.425, SE = 0.074, p < .01). This suggests that nonwhite individuals had 42.5% lower income in 2024 than white individuals, all else being equal.
3. **Gender × Race Interaction**:
   - The interaction term `FEMALE:NONWHITE` is significant ($\beta$ = 0.825, SE = 0.044, p < .01), indicating that the income gap associated with gender in 2024 was mitigated for nonwhite women.
4. **Marital Status**:
   - Being married (`MARRIED`) is associated with a significant increase in income ($\beta$ = 4.608, SE = 0.062, p < .01), suggesting that married individuals in 2024 had 460.8% higher income on average compared to unmarried individuals.
5. **Education**:
   - The coefficient for having a high school diploma or higher (`HIGHSCHOOL`) is significant ($\beta$ = 0.345, SE = 0.074, p < .01), indicating that high school or higher graduates earned 34.5% more than those without a high school diploma in 2024.
6. **Region**:
   - The regional dummy variables reveal significant income disparities across regions:
     - Individuals in Region 2 (`R2`) earned 49.0% less in 2024 than those in Region 1 (baseline) ($\beta$ = -0.490, SE = 0.065, p < .01).
     - Individuals in Region 3 (`R3`) earned 23.7% less than those in Region 1 in 2024 ($\beta$ = -0.237, SE = 0.067, p < .01).
     - Individuals in Region 4 (`R4`) earned 69.0% more than those in Region 1 in 2024 ($\beta$ = 0.690, SE = 0.101, p < .01).
7. **Constant Term**:
   - The intercept is significant ($\beta$ = 1.306, SE = 0.081, p < .01), representing the baseline `log(INCWAGE)` for men, white individuals, unmarried, without a high school diploma, and residing in Region 1 in 2024.

## (17.3) Model Performance:

- **R²**: The model explains 6.8% of the variance in `log(INCWAGE)`.
- **Adjusted R²**: The model explains 6.7% of the variance in `log(INCWAGE)` after accounting for the number of predictors in the model.
- **F-statistic**: The overall model is statistically significant ($F(8; 115827) = 1{,}048.518$, $p < .01$), suggesting that the predictors collectively have a significant effect on income.

## (17.4) Conclusion:

Model 3 highlights significant disparities in income associated with gender, race, marital status, education, and geographic region. This emphasizes the interplay of demographic and regional factors in shaping income distributions.